

FAILURE MODE EFFECTS & CRITICALITY ANALYSIS (FMECA)

Team [ANONYMIZED] — Phase 1 Submission

Independent Submission • IEEE RAS × IEEE AECS Tunisia Section TSYP14 Technical Challenge

FAILURE CASES & MITIGATIONS: THE LIVING MAP

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1. OVERVIEW

The Living Map architecture explicitly anticipates platform loss, environmental degradation, and communication disruption in subterranean environments. Below is the comprehensive failure modes, effects, and criticality analysis (FMECA).

2. FAILURE MODE ANALYSIS MATRIX

ID	Failure Mode	Trigger / Root Cause	Detection Signal	Immediate Autonomous Response	Recovery / Handover Action	Residual Risk
F1	Writer robot destruction or complete power depletion	Structural collapse, battery exhaustion (<5%), motor seizure	ONA detects absence of Writer beacon heartbeat for > 2× broadcast period	Writer fails safely; all prior state exists in deposited beacons	Executor is briefed with partial map up to last beacon; proceeds from last known beacon	Unexplored tunnel volume beyond failure point
F2	Intermediate beacon loss or destruction	Physical debris impact, battery failure, RF shielding	Beacon hop sequence gap in gossip relay; Executor RSSI timeout	Executor switches to dead-reckoning on stored bearing/distance toward next expected beacon	Retraces path to last verified beacon; switches to wall-following until next beacon detected	Increased odometric uncertainty over gap
F3	Odometric drift accumulation in GPS-denied tunnel	Wheel slip on wet rock/debris, gyro drift	Uncertainty radius $\sigma(s) = 0.30 + 0.02s$ exceeds threshold; disparity at beacon landmarks	Robot re-anchors pose upon reaching beacon RSSI peak (known coordinates)	Accumulator reset; corrected coordinates broadcast in updated frame	Small residual local localization error (<0.5 m)
F4	RF packet collision and burst multipath interference	Concurrent broadcast by multiple beacons; rock reflection	CRC-16 checksum failure; duplicate frame counter mismatch	Frame discarded immediately; randomized transmission jitter (0–500 ms)	Gossip relay repeats broadcast over multiple cycles; ALOHA retry	Sub-second transmission latency under heavy interference
F5	ONA to Command Post uplink outage	Satellite fade, LTE antenna damage, terrain obstruction	ONA edge processor detects TCP ACK timeout (> 5 s)	Frames held in persistent local SQLite store-and-forward queue	Uplink re-established: queue flushed in FIFO order; CP displays data age warning	Command Post sees delayed live map during disconnection
F6	Dynamic structural collapse after Writer transit	Secondary mine tremor, falling rock/beam	Executor LiDAR detects occupied cells in previously cleared corridor	Executor halts immediately; transitions to REROUTE state; writes BLOCKED beacon	Evaluates alternate branch from nearest junction; aborts to ONA if blocked	Trapped corridor segment impassable
F7	Hazard volatility / Stale gas information	Gas cloud dissipation or ventilation airflow shift	Confidence metric $c(t) = 2^{-(t-T)/\tau}$ decays below 0.6 (AGING) or 0.25 (STALE)	Executor approaches target in VERIFY state at reduced velocity (0.1 m/s)	Robot re-senses concentration; if clear, updates beacon with cleared status; refreshes timestamp	Transient sensor reading in turbulent air
F8	Primary 2D LiDAR failure or optical obscuration	Dust storm, lens fracture, sensor electronics failure	Scan driver reports timeout or zero-point return across all rays	Robot immediately stops forward transit; drops emergency breadcrumb beacon	Falls back to tactile/ultrasonic obstacle avoidance at creeping speed and returns to entrance	Inability to map fine obstacles
F9	Master clock drift across distributed nodes	Asymmetric oscillator temperature coefficient	Beacon timestamp inconsistency relative to ONA arrival time	ONA stamps frames with arrival time T_{arr} ; computes relative mission delta Δt	Mission clock T_{0} synchronized at Executor briefing prior to tunnel entry	Sub-second timing jitter in confidence decay